

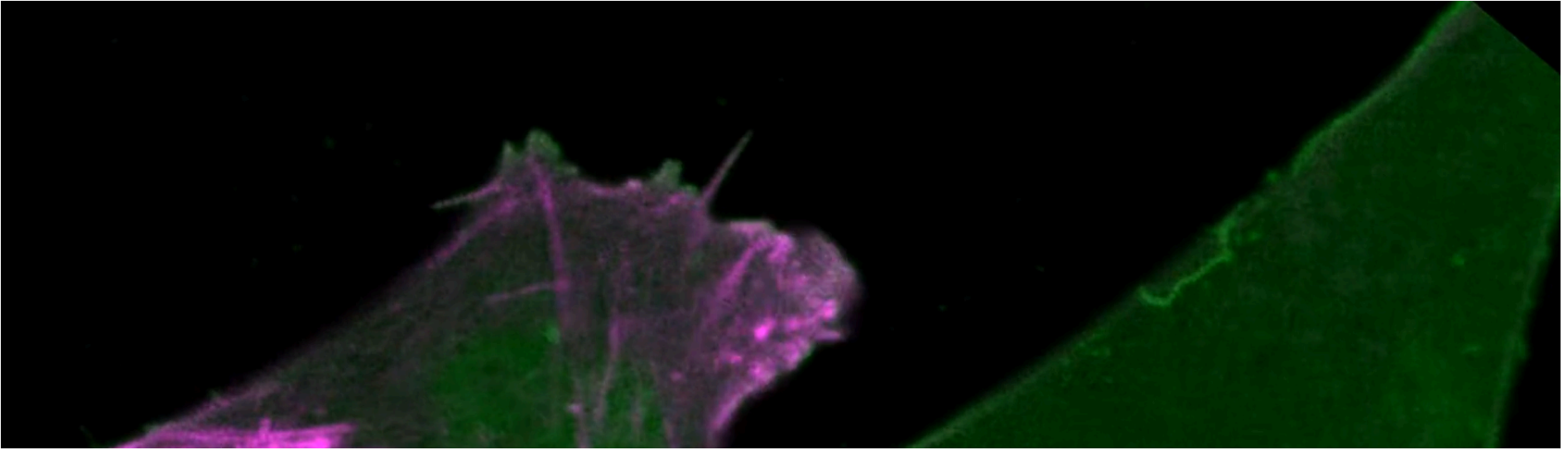


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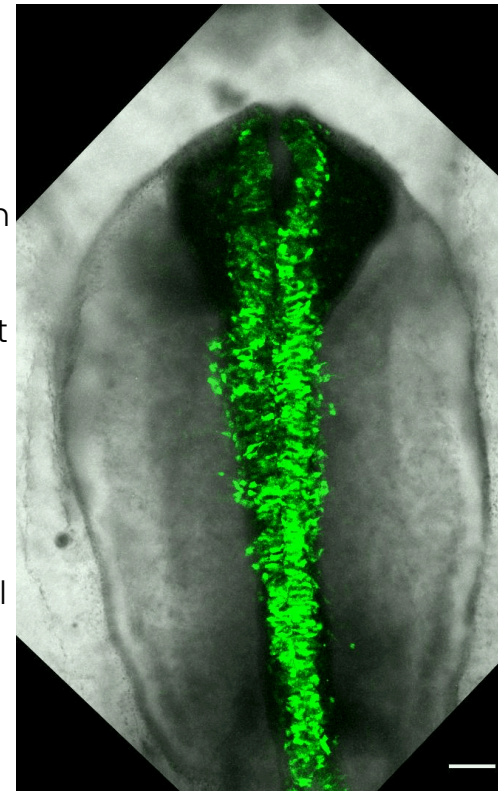
How does membrane organization control epithelial-to-mesenchymal transition and cell migration during development and disease?



EMT and cell migration

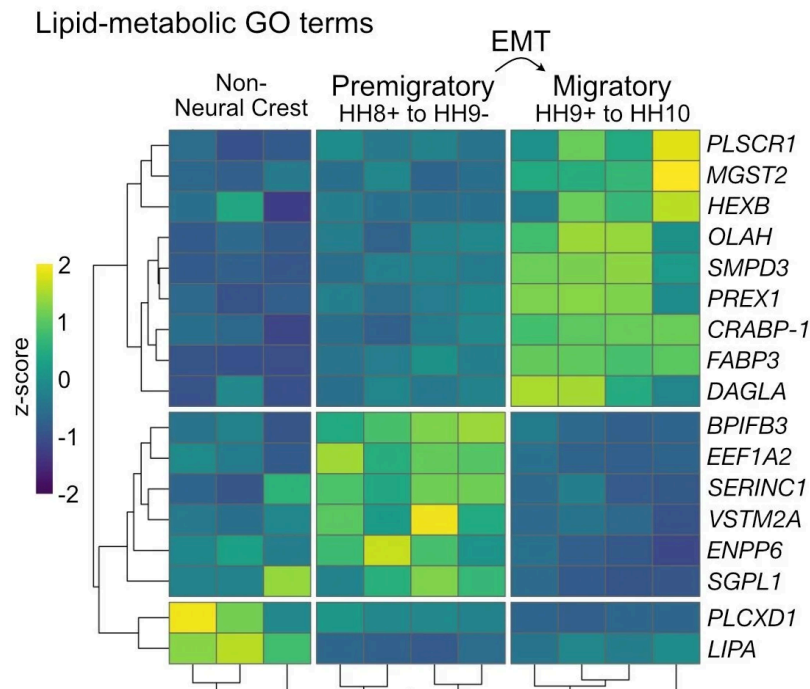
Epithelial-to-mesenchymal transition, or EMT, is a major cellular lifestyle change that results in cells with the ability to migrate long distances in the body. This process is critical for normal embryonic development and for the formation of many organs. However, EMT mechanisms are often co-opted in disease states such as cancer metastasis.

We use the chicken neural crest as a tractable model system to study the mechanisms of EMT and cell migration. The neural crest is an embryonic cell population with remarkable multipotency and



migratory ability enabling it to contribute to diverse organ systems including the craniofacial skeleton and the peripheral nervous system (**Piacentino et al., 2020, *Current Opinion in Cell Biology***). Thus, insights gained from studying neural crest cells have the potential to inform novel therapeutic approaches to combat both congenital disorders and metastatic diseases.

Lipid metabolism controls neural crest EMT



We are beginning to appreciate that EMT is often punctuated by changes in expression of diverse lipid metabolizing enzymes raising the question: what is the function of lipid metabolism during EMT?

To date, we have described the mechanism by which the production of the membrane lipid ceramide by the enzyme nSMase2 is necessary for neural crest EMT (**Piacentino et al., 2022, *PNAS***). Our work shows that ceramide production in the neural crest is necessary for endocytosis which promotes cell

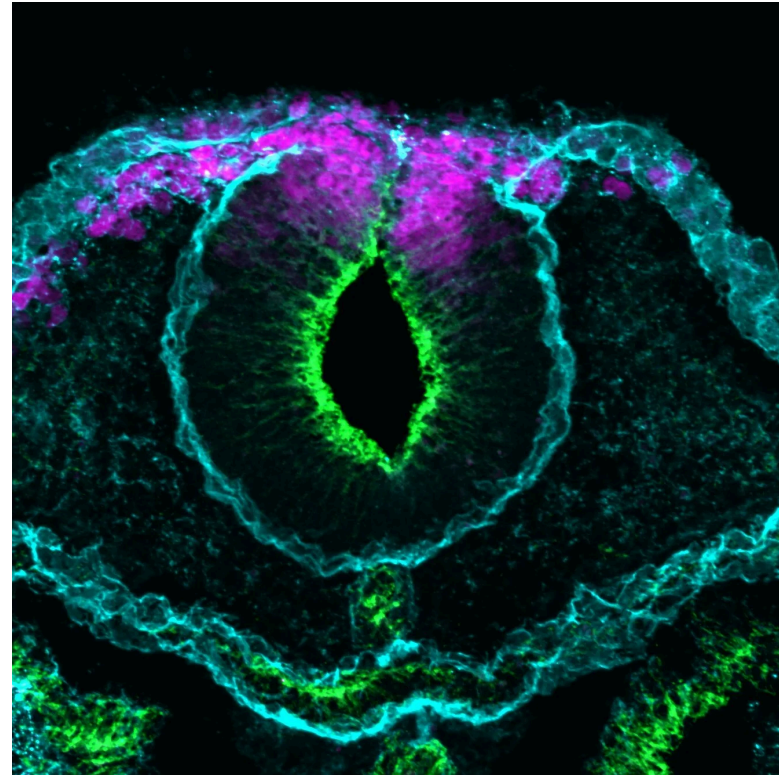


signaling, transcriptional regulation, de-adhesion and delamination.

Using transcriptomics, reporter constructs, and gene editing, we are now deciphering how different lipid metabolizing genes are transcriptionally regulated across embryonic space and time.

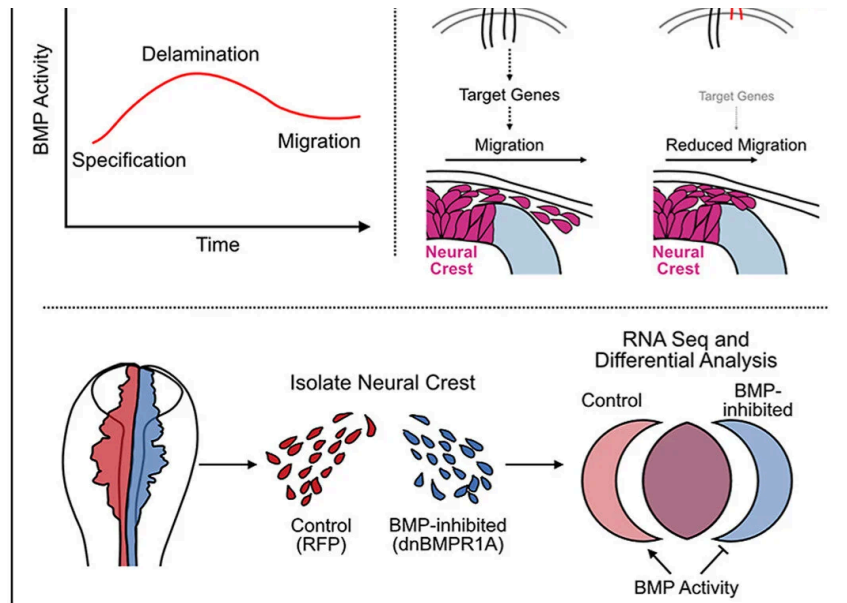
How does membrane biophysics direct cell migration?

Changing lipid metabolism has the potential to modulate the biophysical properties of the plasma membrane. We are interested in studying how temporally-controlled lipid metabolism alters membrane properties such as fluidity/rigidity as well as lateral organization of proteins/lipids. Accordingly, we have found that ceramide production by nSMase2 increases plasma membrane fluidity to influence directional neural crest migration (coming soon to bioRxiv).



How is cell signaling tuned during development?





Cell signaling including the BMP and Wnt pathways play critical and iterative roles during neural crest development; yet, how these signals occur at precise times and levels remains unclear. We have identified a mechanism by which the membrane-bound scaffold protein, CKIP-1, tunes BMP signaling during neural crest induction by controlling its ubiquitin-mediated degradation (**Piacentino** and Bronner, 2018, *PLOS Biology*).

We then found a previously undescribed role for BMP signaling during neural crest migration away from the midbrain. Using transcriptome profiling on sorted neural crest cells, we have identified novel targets of BMP signaling during cranial neural crest migration (**Piacentino** et al., 2021, *Developmental Biology*), and look forward to understanding the functions of these new genes!

Ongoing work in the lab seeks to understand how:

- 1) The structural and signaling properties of lipids regulate neural crest development
- 2) The organization of membrane lipids and proteins tune cell signaling

- 3) Cells alter their biophysical properties to coordinate invasive behaviors like cell migration

Contact the Piacentino Lab

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